21-S2-Q3

$$Q\,2000 \begin{cases} 1200 & 1 \\ 800 & 2 \end{cases}$$

(a) Bayes x_1

(b) Fisher x_2

(c) procedure + metrics → classifiers

Solution (a) Bayes

① we define the discriminant function

$$g_i(x) = p(x|w_i)\,p(w_i)$$

If $g_1(x) > g_2(x)$, classifying x to class 1

If $g_1(x) < g_2(x)$, classifying x to class 2

If $g_1(x) = g_2(x)$, x on the decision boundary

$$p(x|w_i) = \frac{1}{(2\pi)^{d/2}\,|\Sigma_i|^{\frac{1}{2}}}\, e^{-\frac{1}{2}(x-\mu_i)^T \Sigma_i^{-1}(x-\mu_i)}$$

② $$p(x_1|w_1) = \frac{1}{2\pi\,|\Sigma_1|^{\frac{1}{2}}}\, e^{-\frac{1}{2}(x_1-\mu_1)^T \Sigma_1^{-1}(x_1-\mu_1)}$$

$|\Sigma_1| = 0.8972$

$(x_1-\mu_1)^T \Sigma_i^{-1}(x_1-\mu_1) = 1.0721$

$p(x_1|w_1) = 0.09830$

③ $$p(x_1|w_2) = \frac{1}{2\pi\,|\Sigma_2|^{\frac{1}{2}}}\, e^{-\frac{1}{2}(x_1-\mu_2)^T \Sigma_2^{-1}(x_1-\mu_2)}$$

$|\Sigma_2| = 1.3544$

$(x_1-\mu_2)^T \Sigma_2^{-1} (x_1-\mu_2) = 3.8531$

$p(x_1|w_2) = 0.01992$

④ $p(w_1) = \frac{1200}{2000} = 0.6$

$p(w_2) = \frac{800}{2000} = 0.4$

⑤ $g_1(x_1) = p(x_1|w_1) \times p(w_1) = 0.05898$

$g_2(x_1) = p(x_1|w_2) \times p(w_2) = 0.007968$

$g_1(x) > g_2(x_2)$

So, we classify x_1 to class 1

(b) Fisher DLA

① $S_1 = n_1 \hat{\Sigma}_1$

$$= 1200 \times \begin{bmatrix} 1.02 & -0.02 \\ -0.02 & 0.88 \end{bmatrix}$$

$$= \begin{bmatrix} 1224 & -24 \\ -24 & 1056 \end{bmatrix}$$

$S_2 = n_2 \hat{\Sigma}_2$

$$= 800 \times \begin{bmatrix} 1.18 & -0.12 \\ -0.12 & 1.16 \end{bmatrix}$$

$$= \begin{bmatrix} 944 & -96 \\ -96 & 928 \end{bmatrix}$$

② $S_w = S_1 + S_2$

$w = S_w^{-1} (\hat{\mu}_1 - \hat{\mu}_2)$

$$w_0 = -\frac{w^T(\hat{\mu}_1 + \hat{\mu}_2)}{2}$$

we define the discriminant function

$g(x) = w^T x + w_0$

③ $S_w = \begin{bmatrix} 2168 & -120 \\ -120 & 1984 \end{bmatrix}$

$$w = \begin{bmatrix} -0.0008330 \\ -0.001300 \end{bmatrix}$$

$w_0 = 0.001367$

$$g(x) = \begin{bmatrix} -0.0008330 & -0.001300 \end{bmatrix} x + 0.001367$$

for $\hat{\mu}_1 = \begin{bmatrix} -0.51 \\ -0.39 \end{bmatrix}$

$$g(\hat{\mu}_1) = 0.0009318 + 0.001367$$
$$= 0.002299 > 0$$

Thus,

If $g(x) > 0$, classifying x to class 1

If $g(x) < 0$, classifying x to class 2

If $g(x) = 0$, x on the decision boundary

④ For $x_2 = \begin{bmatrix} 2.244 \\ 1.491 \end{bmatrix}$

$$g(x) = -0.003808 + 1.001367$$
$$= -0.002441 < 0$$

So, we classify x_2 to class 2

(c) procedures

① Hold-out, repeated hold-out

② K-fold cross-validation

③ Repeated K-fold cross-validation

④ Leave-one-out

Metrics

① accuracy, error rate

② sensitivity and specificity

③ Precision, recall, F-score

④ ROC curve and AUC